

The Online Disinhibition Effect

Final Report EMPRA FOMO: Formal Modeling of Psychological Theories

im Bachelorstudiengang Psychologie der LMU München

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Introduction

The aim of the following work is the formalization of (parts of) Suler's (2004) Online Disinhibition Effect. This goal involves three steps: (1) Formalization of the existing theoretical framework using a VAST display (Leising et al., 2023), (2) development of a formal mathematical model of the narrative interconnections and (3) evaluation of the developed model through data simulation and hypothesis testing.

Suler (2004) describes six interacting factors (Dissociative Anonymity, Invisibility, Asynchrony, Solipistic Introjection, Dissociative Imagination, Minimization of Status and Authority) believed to cause an effect come to known as online disinhibition, which can result in increased toxic (e.g. cursing, threatening remarks) or benign behaviors (e.g. giving advice and emotional support, self-disclosure) during online presence compared to face-interactions.

Formalization of the Online Disinhibition Effect

In order to achieve the formalization in the course of the seminar, we had to limit our scope to two of the six factors proposed by Suler (2004). Because of this restriction, our model no longer aims to explain the difference between online and offline behavior in general, but rather the difference in occurrence of disinhibited behavior in online contexts with higher vs. lower anonymity and invisibility. Figure 1 presents our final VAST display.

Core Constructs & Proto-Theory

First, dissociative anonymity (A, see Appendix A: Construct Source Table) was divided into several components in order to better explain the factor's effect on a person's behavior. Anonymity (A1) refers to the alteration or concealment of a person's real identity when engaging in online activities. The manifestation of anonymity predicts identity compartmentalization (A2), which describes the

separation of a person's identity into an online and offline segment. In the most extreme cases, people might "convince themselves that those online behaviors 'aren't me at all'" (Suler, 2004, p. 322).

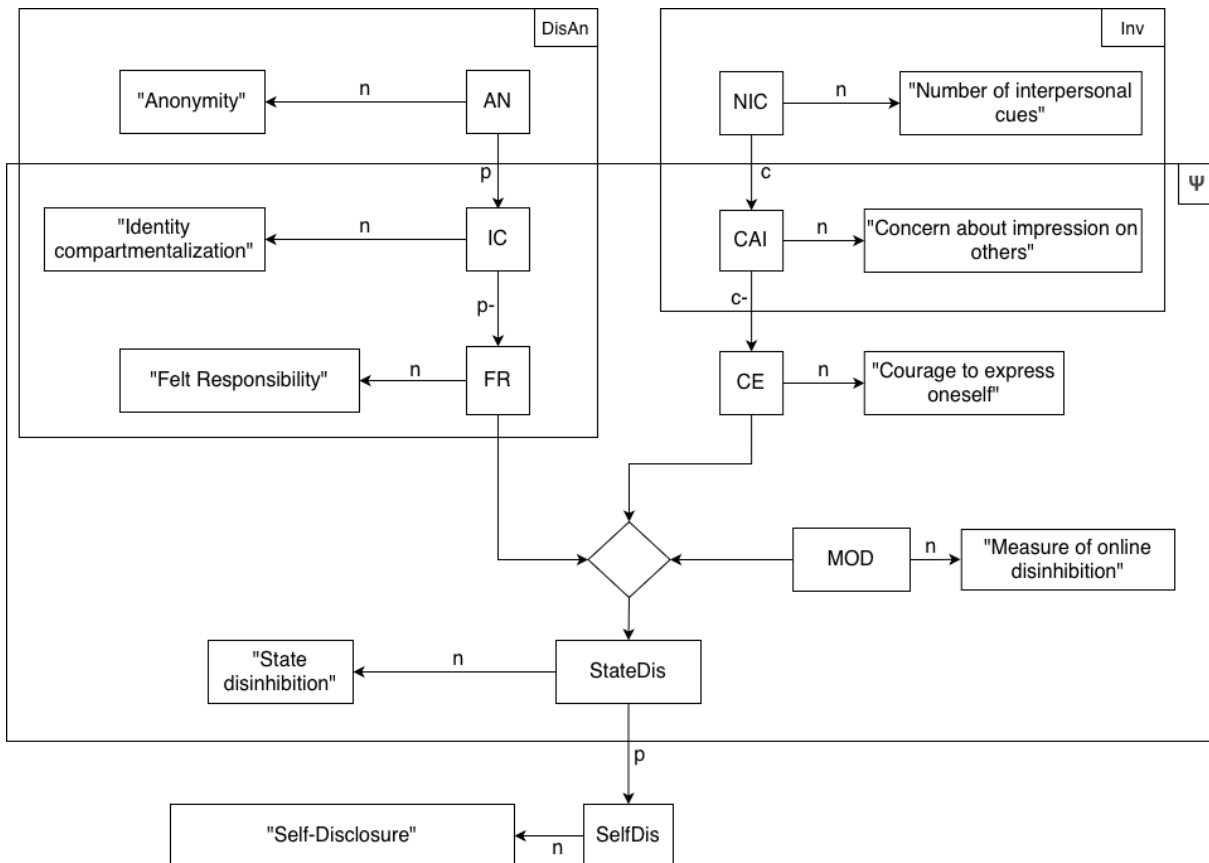


Figure 1: VAST display

The extent of identity compartmentalization in turn is related to the felt responsibility (A3) for a person's online behavior: If people tend to secede these behaviors from their offline identity, they can avert responsibility for them.

With the number of tangible interpersonal cues (B1, e.g. approving/disapproving facial expressions), a person's concern about his or her impression on others increases (B2). Conversely, as invisibility (B) is a common characteristic of online

environments, “[p]eople don’t have to worry about how they look or sound when they type a message. They don’t have to worry about how others look or sound in response to what they say” (Suler, 2004, p. 322). Consequently, depending on the degree of this concern, people feel more or less courage to express themselves (B3).

In accordance with Suler’s (2004) assumptions, we build our model in a way that let dissociative anonymity and invisibility positively interact with each other.

Along with anonymity and invisibility, there are further factors that contribute to the emergence of disinhibited online behavior. Personality variables constitute an important group (Suler, 2004). As a representative example with special relevance to our model, we incorporated the Measure of Online Disinhibition. This instrument allows “assessing self-perceptions of psychological and behavioral change in the online as compared to the offline environment” (Stuart & Scott, 2021, p. 4). Our illustration of differences in disinhibited behavior in online contexts with higher vs. lower anonymity and invisibility therefore includes a trait Online Disinhibition.

The interplay of these three factors explains a more or less pronounced psychological state disinhibition (D), which is expected to reflect in observable online behavior.

During the group stage of the seminar, we specified cursing (as part of toxic disinhibition) as observable outcome of our model. For the individual part, cursing was replaced by self-disclosure, a form of benign disinhibition that refers to the revelation of personal information to others (Lapidot-Lefler & Barak, 2015). Suler (2004) describes toxic and benign behaviors as equally plausible phenomena associated with the Online Disinhibition Effect. Accordingly, self-disclosure should be just as likely an outcome under the conditions of anonymity and invisibility as cursing, and our model should be fully capable of explaining it.

Benevolent Disinhibition in Form of Self-Disclosure: Strength of Evidence

In 2019, Clark-Gordon et al. conducted a meta-analysis ($k = 14$) on the phenomenon of amplified self-disclosure as a result of higher anonymity. While invisibility was not explicitly conceptualized as a separate predictor, their definition of anonymity encompasses invisibility: The construct is divided into discursive anonymity (which means that specific verbal comments cannot be attributed to a specific individual source) and visual anonymity, which refers to the lack of visual representation of a communication partner.

In their study, Clark-Gordon et al. (2019) examine the effect only for the “combined” predictor of anonymity; not two separate main effects for anonymity and invisibility, as would have been ideal for an evaluation of ours. However, the type of anonymity (discursive vs. visual) is included as a moderator in the meta-analysis. Since the distinction is at least acknowledged in terms of moderation, we draw on the results of Clark-Gordon et al. (2019) below to evaluate the empirical strength of our phenomenon.

The analyses overall show a small positive effect ($r = 0.18$) of anonymity (6 studies manipulating visual anonymity, 8 studies manipulating discursive anonymity) on self-disclosure. Further, the conceptualization of visual vs. discursive anonymity did not lead to statistically significant differences, which is compatible with the theoretical assumption that both factors should lead to an increase in self-disclosing behavior.

In sum, the results show that anonymity as well as the presence of nonverbal and parasocial cues impacts self-disclosure in online environments. Concurrently, this relationship is characterized by heterogeneous effect sizes and directions, which could not be explained even by taking several moderators into account.

UTOS Dimensions

In order to conduct a comprehensive evaluation of the phenomenon's robustness, variability across UTOS dimensions (Findley et al., 2021) is examined below.

Units

While Clark–Gordon et al. (2019) do not provide an overview of the composition of samples, the primary studies report predominantly young participants (in many cases university students) from various countries and cultures (e.g. China (Chen et al., 2016), Korea (Kang & Yang, 2004), several english-speaking countries (Hollenbaugh & Everett, 2013 ; Knoll & Bronstein, 2014)).

Treatment

Both anonymity and invisibility are mostly conceptualized as dichotomous variables, which contradicts the recommendation of Clark–Gordon et al. (2019) to “consider anonymity as a continuum rather than binary state” (S. 107).

(Discursive) anonymity is typically manipulated by using random aliases or pseudonyms and, in contrast, by assigning (fictional) identifiers like name, gender or job. (Lapidot–Lefler & Barak, 2015). Invisibility is operationalized as absence or presence of a profile picture (Clark–Gordon et al., 2019) or visibility vs. invisibility of the partners upper body via camera (Lapidot–Lefler & Barak, 2015).

Outcome

Self–Disclosure is often measured using one or more of different tools. One possibility is a textual analysis conducted by an expert, whose task it is to categorize the occurrence of self–disclosing behavior into no, low or high degree of self–disclosure (Lapidot–Lefler & Barak, 2015). Another option is a textual analysis where prespecified expressions (e.g. “I think”) are counted. Their sum serves as

indicator for the person's tendency to self-disclose. In addition to or instead of these, self-reports on self-disclosing behavior are used (Clark-Gordon et al., 2019).

Setting

Reported settings included simulated scenarios in laboratories where participants communicated via chat with other subjects (either based on a given prompt or informally), as well as the administration of short online surveys and subsequent analyses of texts that participants had published on their online profiles.

In sum, even though a considerable number of variations over the UTOS dimensions was tested, the given empirical evidence remains inconclusive due to the specifics of the individual studies considered: It is possible that the research questions investigated are so narrow that they do not rule out spurious variables (Clark-Gordon et al., 2019). Nevertheless, it should be noted that the substantial heterogeneity observed suggests that the examined phenomenon seems to be not generalizable. Based on the current research, increased self-disclosure as a result of higher anonymity and invisibility in online contexts cannot be considered a robust phenomenon.

Development of Formal mathematical Model

All R scripts and additional materials created during the collaborative and individual work can be found in the following public Git repository:

Buss, M., Haas, L., Luchiean, S., & Schlieben, A. (2025). Online Disinhibition Effect: A formal model (Version 1.0.0) [Computer software].

https://github.com/amelieschlieben/FOMO_Eigenleistung

Since we had to drastically limit our theoretical scope already and created our VAST with the mathematical formalization in mind, no further preparations of our model

were necessary for this step. Table 1 displays an overview of all VAST concepts that were reduced to metrics in order to build the formal model.

Table 1

Variable Table

Construct	Short Name	Scale Level	Range/ Values	Anchors
Anonymity	AN	continous	[0;1]	0 = User is fully known during the interaction 1 = Full anonymity where the user remains unknowable during the interaction
Identity Compartment-alization	IC	continous	[0;1]	0 = online- and offline-self are completely identical 1 = feeling like ‘those online behaviors “aren’t me at all.”’ (Suler, 2004)
Felt Responsibility	FR	continous	[0;1]	0 = one feels no responsibility for one’s actions, temporary suspension of moral cognitive processes 1 = maximal feeling of responsibility for one’s actions
Number of Interpersonal Cues	NIC	continous	[0;1]	0 = text-only, minimal interpersonal cues 1 = audio, video and nonverbal feedback, maximal nonverbal, paraverbal and verbal cues
Concern about Impression on Others	CAI	continous	[0;1]	0 = not concerned about impression on others 1 = maximally concerned about impression on others
Courage to Express Oneself	CE	continous	[0;1]	0 = maximal inhibition and fear of expressing oneself; 1 = no fear of expressing oneself

Measure of Online Disinhibition	MOD	continuous	[0;1]	5-point likert scale; 1 = strongly disagree 5 = strongly agree for the items of MOD, high trait disinhibition
State Disinhibition	StateDis	continuous	[0;1]	-0.1 = no inclination to “say and do things in cyberspace that they wouldn’t ordinarily say and do in the face-to-face world” (Suler, 2004) 1.2 = maximal inclination toward such behavior
Self-Disclosure	SelfDis	categorical	[0, 1, 2]	0 = no self-disclosure of biographical information, personal thoughts or emotions, judged by experts 2 = high degree of self-disclosure of biographical information, personal thoughts or emotions, judged by experts (Lapidot-Lefler & Barak, 2015)

Most of the mathematical relationships defined are not directly derived from theory and rather based on heuristic assumptions. While there exists no robust empirical evidence for either of them, there are, in some cases, indications for their reasonableness.

The degree of identity compartmentalization (A2) in our model is determined by the extent of anonymity (A1). We chose the following formula:

$$IC = \frac{AN^2}{AN^2 + (1 - AN^2)}$$

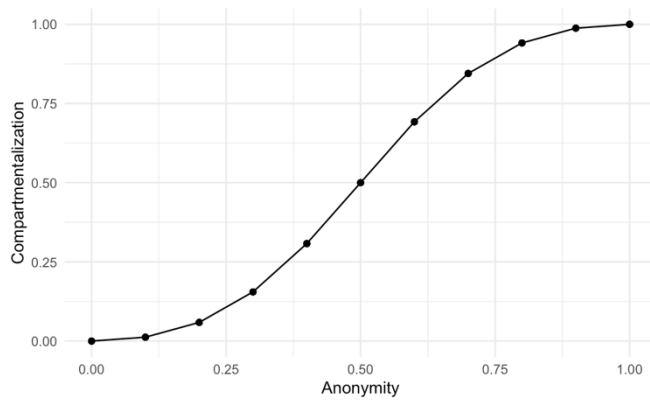


Figure 2

Effect of anonymity on identity compartmentalization

As Figure 2 shows, this produces a positive non-linear relationship.

The S-shape reflects our assumption that a very low level of anonymity does not immediately lead to identity

compartmentalization, as well as a very high level of anonymity does

not have any further noticeable effect on compartmentalization.

The outcome of the IC-function in turn dictates the felt responsibility (A3):

$$FR = base_{resp} * (1 - 0.8 * IC)^3$$

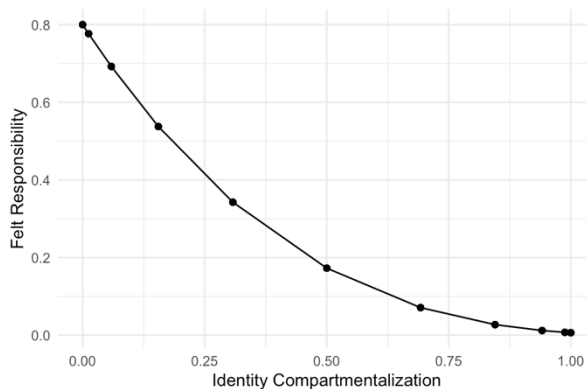


Figure 3

Effect of identity compartmentalization on felt responsibility

baseresp stands for base responsibility, which describes a person's trait accountability and was included to account for inter-individual variation between participants. As observable in Figure 3, the higher identity compartmentalization, the lower the felt responsibility.

As shown in Figure 4, the number of interpersonal cues (B1) can be directly translated in a person's concern about their impression on others (B2).

$$CAI = NIC$$

This concern then causes the person's courage to express oneself (B3): The higher the concern, the lower the courage (see Figure 5).

$$CE = -CAI + 1$$

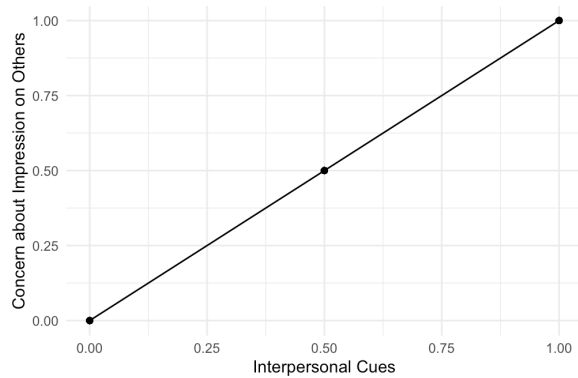


Figure 4

Transformation of interpersonal cues into concern about impression on others

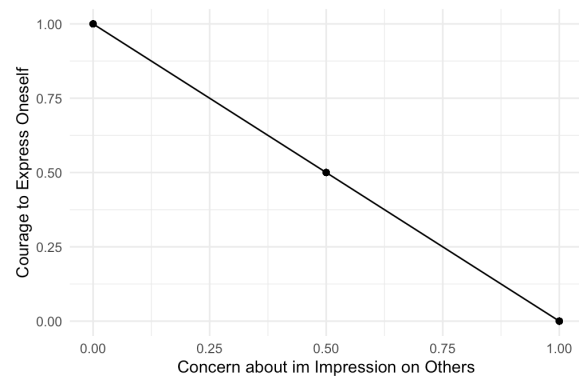


Figure 5

Effect of concern about impression on others on courage to express oneself

State Disinhibition (D) finally results from the interplay of felt responsibility (A3), courage to express oneself (B3) and trait online disinhibition (MOD, C). The mathematical relationship assumed incorporates main effects for these predictors as well as an interaction term for felt responsibility and courage to express oneself:

$$StateDis = 0.2 * MOD - 0.3 * FR + 0.2 * CE - 0.1 * FR * CE$$

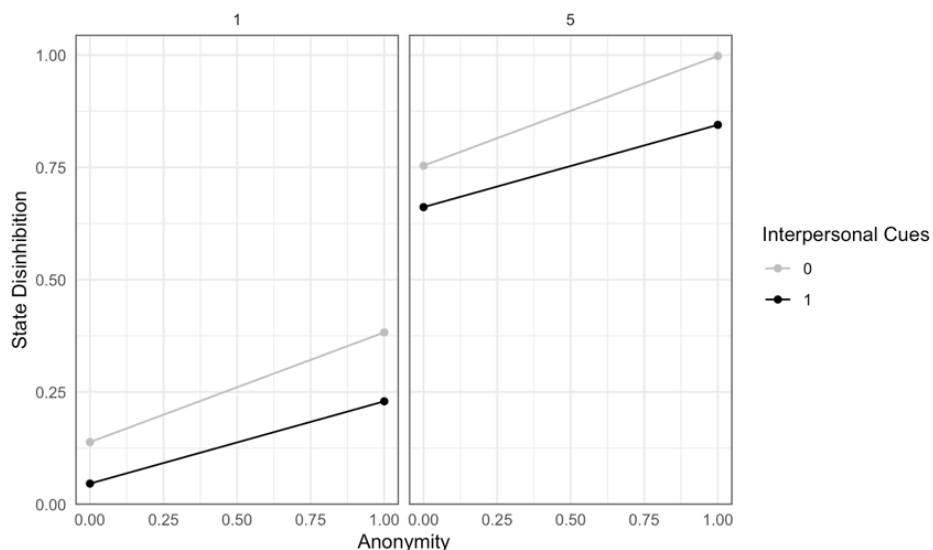


Figure 6

State disinhibition depending on anonymity, interpersonal cues and MOD

Now, noise is added and the produced score is divided into intervals to determine the resulting categorial level of self-disclosure (E). Values in the range $[0; 0.4]$ equate to „no self-disclosure“ in the expert rating, $[0.4; 0.6]$ corresponds to „low self-disclosure“ and $[0.6; 1]$ represents „high self-disclosure“.

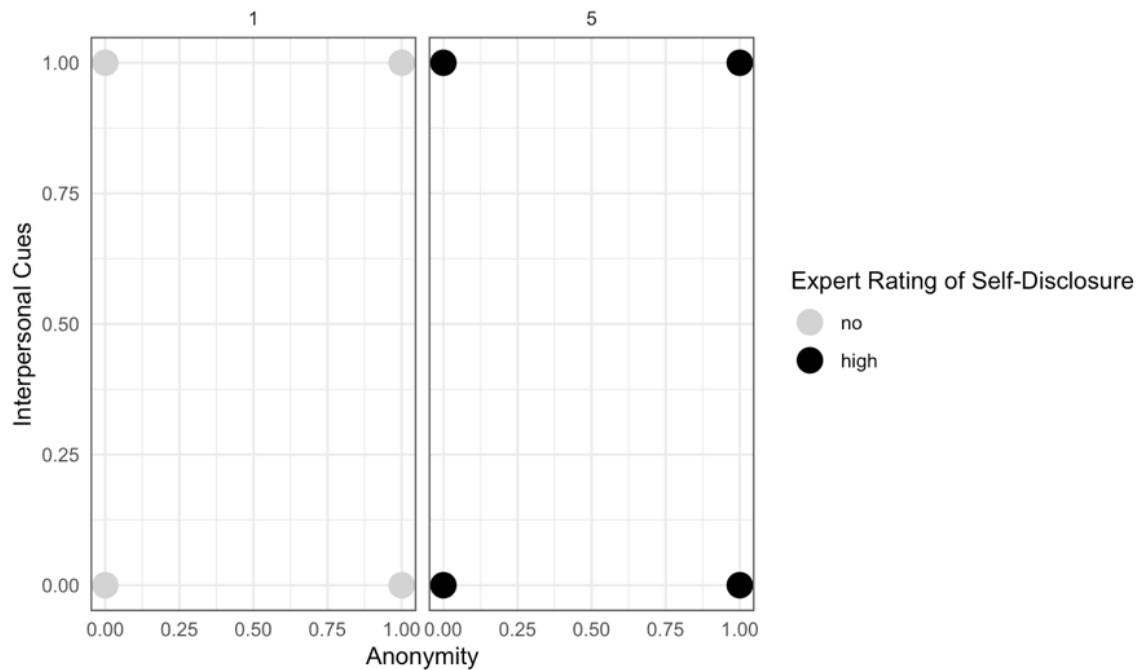


Figure 7
Expert-rated level of self-disclosure depending on anonymity, interpersonal cues and MOD

Evaluation

Data Simulation & Statistical Analysis

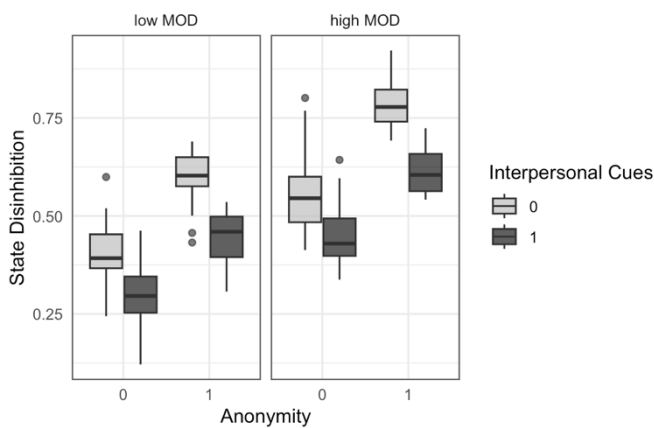


Figure 8
Results ($n=400$) for state disinhibition based on simulated values for anonymity, interpersonal cues, MOD and *baseresp*

Values ($n = 400$) for anonymity, interpersonal cues, MOD (from $\text{beta}(4, 4)$) and *baseresp* (from $\text{beta}(4, 1.5)$) were simulated and tested against prespecified hypotheses. The results of the simulation are shown in Figures 8 and 9.

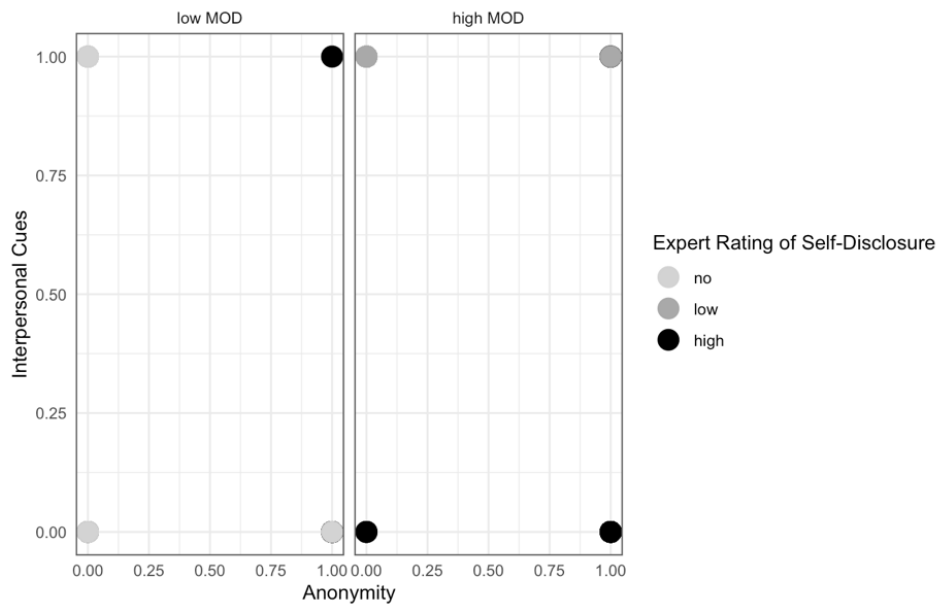


Figure 8

Results (n=400) for expert-rated level of self-disclosure based on simulated values for anonymity, interpersonal cues, MOD and *baseresp*

Hypotheses

Our hypotheses followed directly from the theoretical assumptions by Suler (2004) on which we based our model. We hypothesized that there would be (1) a positive main effect for anonymity on the observable level of self-disclosure, (2) a negative main effect for the number of interpersonal cues on the observable level of self-disclosure and (3) a significant interaction between anonymity and the number of interpersonal cues regarding the observable level of self-disclosure.

Results

Utilizing an ordinal logistic regression from the R-package MASS (Venables & Ripley, 2002), odds ratios were calculated. One-sided t-tests were performed for hypotheses 1 and 2 and a two-sided t-test for hypothesis 3.

Table 2*Regression Coefficients, Confidence Intervals and Results of t-tests*

Predictor	Estimates	CI	p-Value
anonymity	14.13	[7.68; 26.76]	4.031159e-17
number of interpersonal cues	0.29	[0.16; 0.49]	4.837207e-06
anonymity * number of interpersonal cues	0.44	[0.19; 0.98]	4.470472e-02

Our results are consistent with the hypotheses. Odds for the next higher level of self-disclosure increase significantly by the factor 14.13 in completely anonymous online contexts compared to non-anonymous contexts. Similarly, odds decrease significantly by the factor 0.29 if online interactions are highly visible compared to the lowest possible level of visible cues. It is worth noting that the estimate for anonymity is very high overall, and particularly in comparison to that of interpersonal cues. In theory, these effects should not differ so greatly and it is not clear which part of our model caused this.

Lastly, the interaction between anonymity and interpersonal cues also turned out to be significant. Given the strong result of anonymity, the direction of this interaction is somewhat non-intuitive and hard to explain: When an online context is both highly anonymous and characterized by a high number of interpersonal cues, odds for the next higher level of self-disclosure decrease. Hence, visibility seems to be the more important factor in our model, even though the small effect size seems to suggest otherwise.

Discussion and Meta-Reflection

Suler's (2004) framework for the Online Disinhibition Effect was much more extensive than we had initially anticipated. Although we imposed significant theoretical constraints from the start (two instead of six possible predictors), in hindsight we still did not narrow our scope enough to allow us to address each of the individual variables within the formalization steps in a satisfying way.

In addition, the original literature was only of a very limited use, since Suler's (2004) original paper seems to be based on heuristic reasoning rather than on empirical findings. Consequently, terms are used somewhat imprecisely and the search of literature that could provide hints for the mathematical formalization proved especially difficult. As a result, we were largely left on our own both in terms of conceptual development during the creation of the VAST and also when it came to the actual mathematical formalization. Making so many decisions mostly on our own often lead to uncertainty and swallowed a lot of our resources.

Still, despite this limitations, we were able to complete all steps of the formalization process and finally even produced acceptable results. Accordingly, I consider this semester's seminar and the development of our model a complete success.

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Appendix A: Construct Source Table

Reference	Quote	Extracted element	Short name	ID
Suler (2004)	<i>"When people have the opportunity to separate their actions online from their in-person lifestyle and identity, they feel less vulnerable about self-disclosing and acting out. [...] The online self becomes a compart-mentalized self."</i>	Dissociative Anonymity	DisAn	A
Suler (2004)	<i>"[...] for the most part others only know what a person tells them. If so desired, people can hide some or all of their identity. They also can alter their identities."</i>	Anonymity	AN	A1
Suler (2004)	<i>"The online self becomes a compartmentalized self."</i>	Identity Compartmentalization	IC	A2
Suler (2004)	<i>"In the case of expressed hostilities or other deviant actions, the person can avert responsibility for those behaviors"</i>	Felt Responsibility	FR	A3
Suler (2004)	<i>"In many online environments, especially those that are text-driven, people cannot see each other. When people visit web sites, message boards, and even some chat rooms, other people may not even know they are present at all"</i>	Invisibility	Inv	B
Suler (2004)	<i>"Avoiding eye contact and face-to-face visibility disinhibits people. Text communication offers a built-in opportunity to keep one's eyes averted."</i>	Number of Interpersonal Cues	NIC	B1
Suler (2004)	<i>"People don't have to worry about how they look or sound when they type a message. They don't have to worry about how others look or sound in response to what they say. Seeing a frown, a shaking head, a sigh, a bored expression,</i>	Concern about Impression on others	CAI	B2

	<i>and many other subtle and not so subtle signs of disapproval or indifference can inhibit what people are willing to express."</i>			
Suler (2004)	<i>"This invisibility gives people the courage to go places and do things that they otherwise wouldn't"</i>	Courage to Express Oneself	CE	B3
Stuart & Scott (2021)	<i>"[...] the Measure of Online Disinhibition (MOD), was developed as a comprehensive and valence free instrument for assessing self-perceptions of psychological and behavioral change in the online as compared to the offline environment."</i>	Measure of Online Disinhibition	MOD	C
Suler, J. (2004)	<i>"people say and do things in cyberspace that they wouldn't ordinarily say and do in the face-to-face world. They loosen up, feel less restrained, and express themselves more openly."</i>	State Disinhibition	StateDis	D
Lapidot-Lefler & Barak (2015)	<i>"A prominent example of personal ramifications of benign online disinhibition is self-disclosure, defined as revealing personal information to others."</i>	Self-Disclosure	SelfDis	E

Appendix B: Statement on the use of generative AI and AI-assisted technologies in the writing process

In writing this paper, I have used the following tools: ChatGPT, Elicit, DeepL, Perplexity, Semantic Scholar. I used it for the following purpose/s:

- ☒ improving linguistic quality and readability
- ☐ writing assistance e.g. for generating table of contents, outlines, first sentences and paragraphs
- ☐ finding relevant citations (correctly marked as such)
- ☒ finding relevant scientific sources (correctly marked as such)
- ☐ translation of citations or text passages (correctly marked as such, e.g. "translated with DeepL")
- ☐ summarizing information
- ☐ transcribing assistance from audio or video
- ☒ generating or optimizing programming code (e.g. R- or Python-Code)
- ☒ researching terms
- ☐ generating term definitions
- ☐ other [please specify]:

After using this tool or service, I reviewed and edited the content as needed and I take full responsibility for the content of this paper. I certify that this paper does not contain lengthy passages (e.g. summary/abstract, or full paragraphs) of purely AI-generated text.

Appendix C: Declaration of Originality

Hiermit erkläre ich, dass ich die vorliegende Arbeit selbstständig und ohne fremde Hilfe verfasst habe. Ich habe keine anderen als die angegebenen Quellen und Hilfsmittel benutzt und alle wörtlich oder sinngemäß übernommenen Stellen als solche kenntlich gemacht.

Haar, 17.03.2026
Ort, Datum, Unterschrift

